Kirchhoff Laws

Experiment 8 • PHYS 210 X01A

David Dreyer

Partner [enter] Lab date [enter] Procedure version [confirm]

PRELAB DRAFT — complete the highlighted fields and measured data before submitting.

Objective

To calculate branch currents using Kirchhoff's laws and Cramer's rule, compare them with currents obtained from measured resistor voltages, and check whether the measured current and voltage sums are consistent with zero.

Theory

At a junction, charge does not accumulate, so the signed currents sum to zero. Around a closed loop, the signed voltage changes sum to zero because energy is conserved. For a resistor, Ohm's law connects its signed voltage drop U, resistance R and current I.

$$\Sigma I = 0; \quad \Sigma \Delta V = 0; \quad I = U/R$$

I is current in A, U and ΔV are voltages in V, and R is resistance in Ω. A negative current means flow opposite the arrow assigned in the circuit sketch. Absolute uncertainties are written ΔR, ΔU and ΔI.

$$\Delta I = \Delta U/R + |U|\Delta R/R^2$$

For a sum or difference, we add the absolute uncertainties. Currents are compared by checking their difference against the combined uncertainty. A Kirchhoff sum is consistent with zero when its uncertainty interval includes zero.

Apparatus and procedure

[Confirm apparatus: two DC supplies, DMM model and ranges, resistor values, mounting board, leads and clips. State the uncertainty method actually used.]

[Prepared procedure — retain only after confirming it describes the lab: We measured the isolated resistors with the power off. We assembled each assigned circuit and checked the source polarities. We recorded the actual source voltages and signed voltage across each resistor with uncertainty. We used the measured resistances and source voltages in Cramer's rule, then compared the calculated currents with U/R. Finally, we checked the measured junction and loop sums.]

[Insert the actual labelled circuit sketches with current arrows, nodes, loop paths and source polarities. Use the guide's sign convention or revise every equation consistently.]

Data and observations

Use signed voltages as defined in the circuit sketches. Blank cells require actual readings. Remove an unassigned circuit only after confirming the required procedure.

Circuit 1

Resistor	Nominal Ω	Measured Ω	± ΔR Ω
R1	220		
R2	100		
R3	100		
R4	470		

Quantity	Signed voltage V	± ΔV V
E1		
E2		
U1		
U2		
U3		
U4		

Circuit 2

Resistor	Nominal Ω	Measured Ω	± ΔR Ω
R1	220		
R2	100		
R3	470		

Quantity	Signed voltage V	± ΔV V
E1		
E2		
U1		
U2		
U3		

[Record fluctuations, polarity checks, rewiring, actual apparatus and deviations from the planned procedure. Attach original data.]

Calculations and current comparisons

The following equations apply to the full manual's circuits, using all branch arrows from a to b. If the simplified circuit was assigned, replace these with equations derived from its actual sketch.

Circuit 1

$J_L + J_M + J_R = 0; \quad (R_1 + R_3)J_L - R_2 J_M = -E_1; \quad R_2 J_M - R_4 J_R = -E_2$

Circuit 2

$J_1 + J_2 + J_3 = 0; \quad R_1 J_1 - R_2 J_2 = -E_1; \quad R_2 J_2 - R_3 J_3 = +E_2$

[Show the measured-value coefficient matrix, right-hand-side vector, D and each replaced-column determinant. Calculate each branch current as Dk/D. Include at least one determinant expanded with numerical values and units.]

Sample experimental current: $I = [U]/[R] = [\ ]\ A = [\ ]\ mA$

$\Delta I = [\Delta U]/[R] + |[U]|[\Delta R]/[R]^2 = [\ ]\ mA$

Theoretical uncertainty convention [confirm; the simplified handout treats theory as exact]. Compare $|I_{exp} - I_{th}|$ with $\Delta I_{exp} + \Delta I_{th}$ and state whether each pair is consistent.

Circuit / resistor	Iexp ± ΔI mA	Ith mA	Difference mA	Consistent
1 / R1				
1 / R2				
1 / R3				
1 / R4				
2 / R1				
2 / R2				
2 / R3				

[For circuit 1, also compare the separately measured currents through R1 and R3. They are in series and should agree within combined uncertainty.]

Kirchhoff checks and discussion

Use the experimental U/R currents for KCL and the measured voltages for KVL. Add the absolute uncertainties of every term in each sum.

Check	S	ΔS	Zero in interval
C1 node a (mA)			
C1 node b (mA)			
C1 left–middle (V)			
C1 middle–right (V)			
C1 left–right (V)			
C2 node a (mA)			
C2 node b (mA)			
C2 top–middle (V)			
C2 middle–bottom (V)			
C2 top–bottom (V)			

[Show one substituted junction sum, one substituted loop sum and their uncertainty sums. Include all required loop equations and report any exact boundary cases.]

Discussion

[State which current comparisons agreed, citing the numerical differences and uncertainties. Discuss whether each junction and loop sum included zero. Explain any disagreement without changing or discarding the recorded readings.]

[Discuss two specific possible limitations supported by your observations. For example, contact resistance adds unmodelled voltage drops; supply drift changes voltages between readings; heating changes resistance; meter loading changes the circuit slightly. Explain the likely effect and how you would reduce it.]

Conclusion

[Give the measured branch-current results with uncertainties, state whether they agreed with the predictions, and state whether KCL and KVL were verified within uncertainty for each assigned circuit.]

References

Camosun College, PHYS 210 Lab Manual Complete 2024, Experiment 8, L8-1 to L8-5 (PDF pp. 105–109); report format and uncertainty rules in the introduction. Separate course handout: PHYS 210/295 Kirchhoff's Rules Lab — Simplified Procedure (1 page).